Aerospace Manufacturing Core
Study Guide

"AMCSG-202"

Tools & Processes

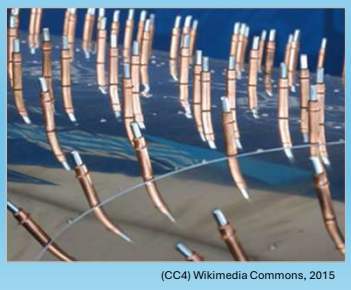
Fasteners

Fasteners

Text from Google AI

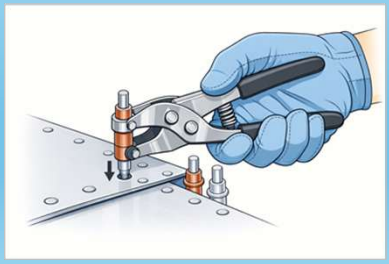
Cleco Temporary Fasteners

- They hold materials during drilling & assembly
- Types: pin, wing nut, nut-power, nut-less-power
- Pin Cleco's are installed with Cleco pliers
- Wing nut Clecos by hand
- Power Clecos by power tool



(CC4) Wikimedia Commons, 2015

(Ad) Pan American Tool Corp, 2026



ChatGPT / Matt Pierce, 2026

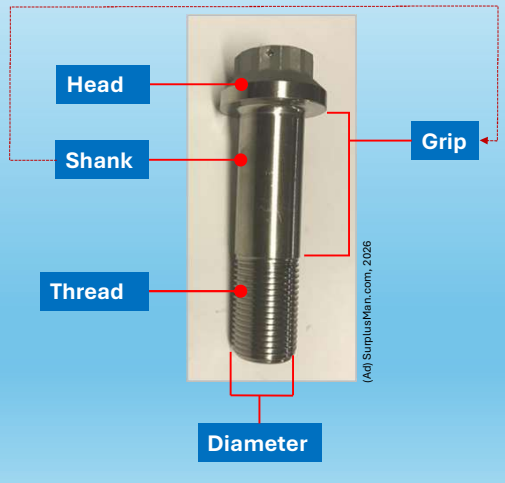
Image Credits:

- Google "Cleco fasteners" = (CC4) - https://commons.wikimedia.org/wiki/File:Cleko.jpg
- Pan American Tool Corp = (Ad) - https://www.panamericantool.com/cleco-fasteners.html?srsltid=AfmBOopSK4AwtAP1zROrgsU2a5QdbdzeT_PqdszlTBGC_92gi_Lq8wEK
- Image of hand installing fastener: ChatGPT: "picture using cleco pliers to place a cleco clamp on aircraft surface" + "make it into a single photo with gloved hand, training-manual style"

Bolts

Threaded fasteners are used for high strength

- Components:



- Types:

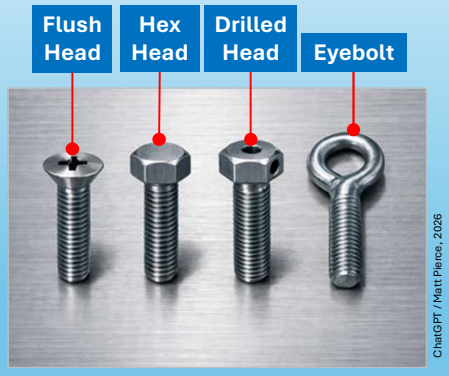
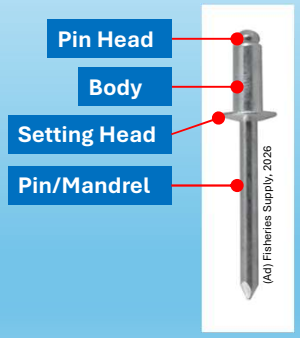


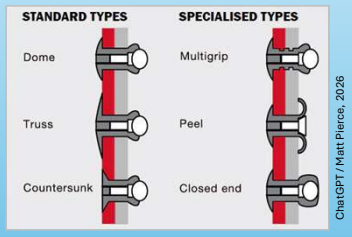
Image Credits:

- Public domain image of “BACB30le12hu27” Boeing Aviation Grade bolt from surplusman.com: https://surplusman.com/item/bacb30le12hu27-boeing-aviation-grade-bolts/9693/
- ChatGPT for bolt head types: “make a picture show several bolt types: flush head, protruding hex head, protruding drilled head, and eyebolt. Remove text labels.”

Rivets

- Lighter than a bolt
- Permanent





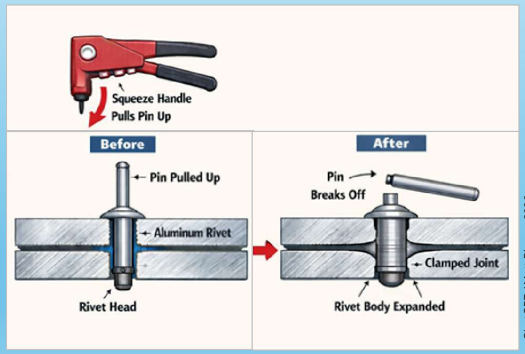


Image Credits:

- Rivet Picture: From Fisheries Supply (sales website): https://encrypted-tbn3.gstatic.com/shopping?q=tbn:ANd9GcRURf0_XjySokKAN7xOMzD0x5XJtuEGAinThvon0nbvy_u4zXJ2o5RO4dk1SZXg1kgJp1Ixcqc_qJO_bzTu_m8W8mOs4CVh-BBIwz4xJ1uek2RheRYr1SrS

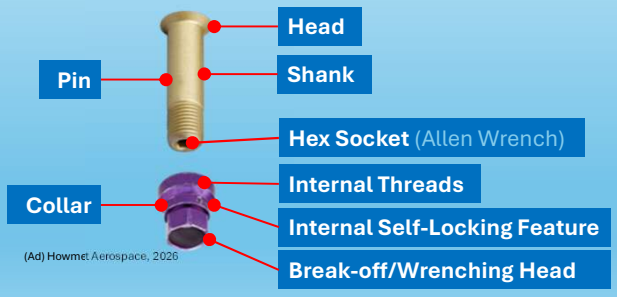
- Rivet Process Diagram: ChatGPT: “make a picture showing how a rivet gun works”

- Rivet Types Picture: ChatGPT: “make a diagram showing how a simple aluminum rivet gun works. Show cross section of two pieces of aluminum. Rivet head on the lower side of two pieces. Rivet body in hole between the two pieces of aluminum. Rivet pin/mandrel above the two pieces. simple red hand held rivet gun attached at the top of the rivet pin/mandrel. show how the rivet pin/mandrel will snap off once the river gun is squeezed.”
  + “remove labels "upper aluminum piece", "lower aluminum piece". Rename "mandrel" to "pin". Use only one arrow at "rivet body expanded". Only use one label (not two) for "Mandrel breaks off". Remove the yellow flash callout for "Mandrel Snaps Off!" Remove title "How a rivet gun works“

+ “remove label "Lower Aluminum Piece". Make the broken off "Pin" longer in the after section, and make it horizontal.”

Hi-Lok Fasteners

- **Lighter** than a bolt
- **Stronger** than a rivet
- **Best** of both



(Ad) Howmet Aerospace, 2026

ChatGPT / Matt Pierce, 2026

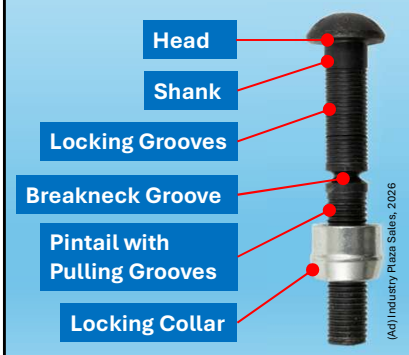
Image Credits:

- Sample Hi-Loks (Veri-Lite brand) for sale on "Howmet Aerospace" site: https://www.google.com/imgres?q=hi-lok%20fastener&imgurl=https%3A%2F%2Fwww.howmet.com%2Fwp-content%2Fuploads%2Fsites%2F3%2F2023%2F05%2FVeri-lite_1.png%3Fw%3D1024&imgrefurl=https%3A%2F%2Fwww.howmet.com%2Fproducts%2Faerospace%2Fpin-systems%2Fveri-lite-pin-fastening-system%2F&docid=7lnd3y8wNJspLM&tbnid=QHe65QBP0_EoLM&vet=12ahUKEwiDyPLQxr6SAxXFEDQIHW5uH-UQM3oECBMQAA..i&w=1023&h=456&hcb=2&ved=2ahUKEwiDyPLQxr6SAxXFEDQIHW5uH-UQM3oECBMQAA

- Installation picture = ChatGPT = "make a picture showing installation of hi-lok fasteners onto aircraft surface using a universal tool pneumatic tool"
  + "remove the cleco fasteners. show the hi-lok fasteners half-way in the hole."
  + "make the pneumatic tool straight. zoom in on the hi-lok fastener after installation with the tool head removed a few inches"

Lockbolt Fasteners

- Stronger than a bolt & nut
- Like a rivet



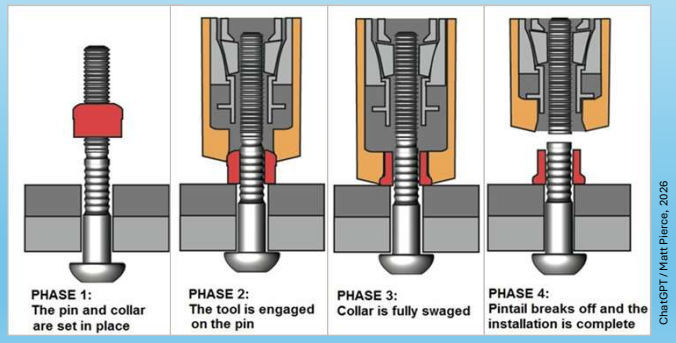


Image Credits:

- Lockbolt Picture: From Industry Plaza Sales website: https://www.industry-plaza.com/lockbolts-huck-c50l-p158808.html
- Lockbolt Process Diagram: ChatGPT: “make a diagram showing how lockbolt fasteners work.” Took the third one from list. First two sets of diagrams not as good.

Text from Google AI

“NAS 523” Fastener Standards

NW Quadrant:
- Engineering Code
- Type/Material

NE Quadrant:
- X=Diamer in 1/32”
- N/F = Location Mfg Head
  (Near or Far Side)

ChatGPT / Matt Pierce, 2026

SW Quadrant:
- D = Dimple
- X = # Sheets to dimple
- C = Countersink

SE Quadrant:
- X = Rivet Length 1/16”
- W = Spot Weld

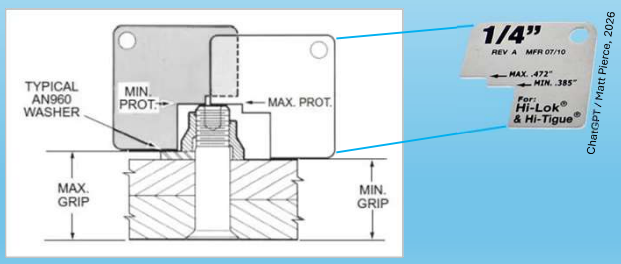
- Google AI: “fastener standards in aircraft manufacturing” for general information

Image Credits:
- Image via ChatGPT: “make a photo of an aircraft manufacturing fastener with example nas 523 quandrant markings“
  + “rename label "A1" to "NW". rename label "B2" to "NE". rename label "C3" to "SW". rename label "D4" to "SE".”

Text from ChatGPT

Inspecting Fasteners: Pin Protrusion (Height)

- Inspecting “Pin Protrusion”
  - Gauge is thin metal card with two-step cutout
  - NoGo at Min Prot = PASS for too small
  - Go at Max Prot = PASS for too big



ChatGPT / Matt Pierce, 2026

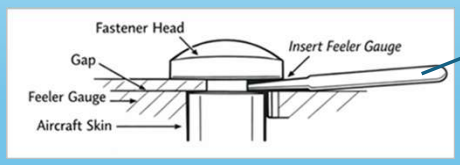
Pin Protrusion Inspection: ChatGPT: “pin protrusion gage thin metal card 1/4“ generates 4 pictures. Using #2 (photo) and #3 (usage diagram).

Text from ChatGPT

Inspecting Fasteners: “Gap” Size

- Starting with smallest feeler gauge…
- Try one by one until not slide freely in
  (goal = find which gauge not slide)
- Gap tolerance is Summative
- Compare NoGo size to engineering spec

(Ad) Ebay For Sale, 2026



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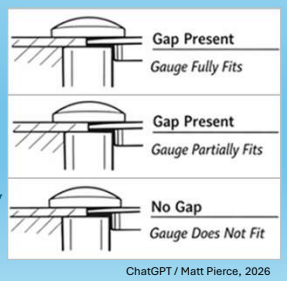


ChatGPT / Matt Pierce, 2026

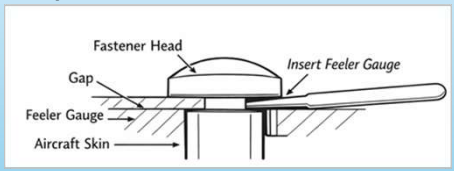
Image Credits:
Picture of Feeler Guage = Ebay for sale picture:
“https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.ebay.com%2Fitm%2F263520840477&psig=AOvVaw0wcBatFQ1RXI_CCPuCQGWl&ust=1770350044353000&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxqFwoTCJD085O2w5IDFQAAAAAdAAAAABAV”

Inspection Steps: ChatGPT: “diagram showing how to inspect fastener gap using feeler gauge. ”

Text from ChatGPT

Inspecting Fasteners: 3-way Contact

- Inspect Contact between Head, Collar, and Structure



ChatGPT / Matt Pierce, 2026

- Start from gap found, then test around fastener until hit at least one point of contact

ChatGPT / Matt Pierce, 2026

Same ChatGPT image as prior slide…just clipped out different pieces for this presentation

Text from ChatGPT

Inspecting Rivets: Diameter & Thickness

To inspect rivet heads:

Use a “spoon gauge”

ChatGPT / Matt Pierce, 2026

...or an “H-gauge”

Image Credits:

- Spoon guage picture for sale at Aircraft Spruce and Specialty: https://encrypted-tbn1.gstatic.com/shopping?q=tbn:ANd9GcRNGHpHs2lFiL8CEYaxgeLJ9edRjc35gAI9kAFvs-HLWRHFw3mFd8HZwz9t5i5axwtov4cXAn8YgN3j-vriDmbtvzWmc2mefg

- H-Gauge picture for sale at VB Industrial Supply: https://encrypted-tbn3.gstatic.com/shopping?q=tbn:ANd9GcSOZm9C-LPr3tfdqzN0fNwlZVm9uctc5rZM0-b0XNONCDIdaaILsY45VJyn4pMXo3Hco9rRI6WiG3amUIu9GI_ktcjArFOv_0ITUeijP4dlGMMWgEP9ve9suv0D

- Usage diagram: ChatGPT: “photo showing inspect rivet with a shop head rivet gauge set. Make it inside fuselage / wing bay realistic with several installed solid rivets. A shop head rivet gauge set (thin metal gauges with full circular cutout).
  A hand holding the gauge flat against the surface, sliding the correct gauge over the shop head

The gauge clearly demonstrating:
- 1/4" diameter requirement (shop just fits inside the gauge opening)
Neutral lighting, realistic materials, no exaggerated colors“

+ “Make the gauge laying down flat around the rivet head. Make the gauge 3x longer with the hand further to the right and the hole further to the left”
+ “Lay the gauge down flat on the surface.”
+ “Make the hole smaller and the rivet head snuggly fit. Make the gauge narrower and longer. Make the gauge bend up 45 degrees from about its midpoint. Make the bent part 2x longer.”

- H-Gauge:

Q&A

Practice Q&A's (Q1-Q9)

Q1: What is the most famous brand of temporary fasteners?

A: Cleco

Q2: Name of a common threaded fastener used for high strength?

A: Bolt

Q3: Name of the thickness two overlapping materials make when fastened together?

A: Grip height

Q4: How do you measure the grip of a threaded protruding head fastener?

A: From the bottom of head to top of threads

Q5: How do you measure the grip of a threaded flush head fastener?

A: From the top of the head to top of threads

Q6: What are bolts comprised of?

A: Bolt and nut

Q7: Are bolts permanent fasteners?

A: No, they can be removed

Q8: What is the most common fastener type in aerospace manufacturing?

A: Rivets...look at the side of any plane :)

Q9: Are rivets permanent fasteners?

A: Yes

Practice Q&A's (Q10-Q19)

Q10: What is an advantage to rivets?

A: Lighter than bolts

Q11: What are rivets comprised of?

A: Pin (or mandrel) and Body

Q12: Why is a Hi-Lok fastener the best of both a bolt and a rivet?

A: Lighter than a bolt, stronger than a rivets

Q13: What are Hi-Loks comprised of?

A: Pin and collar

Q14: Are Hi-Lok fasteners permanent?

A: Yes

Q15: What is an advantage of Lockbolts?

A: Stronger than bolt & nut. Locking collar will not rattle loose.

Q16: What are Lockbolts comprised of?

A: Bolt, breakaway pintail, and locking collar

Q17: Are Lockbolts permanent?

A: Yes, the locking collar is swaged online

Q18: What does NAS stand for (of NAS 523)?

A: National Aerospace Standard

Q19: What does the NW quadrant of NAS contain?

A: Fastener Type (engineering code)

Practice Q&A's (Q20-Q28)

Q20: What does the SQ quadrant of NAS contain?

A: (D)imple or (C)ountersink or blank

Q21: What does the NE quadrant of NAS contain?

A: Diameter in 1/32ths of an inch; Near/Far side

Q22: What does the SE quadrant of NAS contain?

A: Length in 1/16ths of an inch; spot weld

Q23: What is the purpose of a feeler gauge?

A: Find first gap size to fail

Q24: How do you use a feeler gauge?

A: Start with thinnest, go up until one not fit

Q25: Name of the distance across a bolt, side-to-side, thru the middle?

A: Diameter

Q26: Does the diameter of a fastener dictate its tolerance?

A: Yes. Larger diameter = larger tolerance

Q27: What is the most common cause of rivet defects?

A: Human error: improper technique or carelessness

Q28: Name both heads of a rivet?

A: 1. Manufactured head 2. Bucked (or driven) head

Practice Q&A's (Q29-Q32)

Q29: Using a feeler guage, what percent gap around a manufactured flush rivet head is acceptable?

A: No more than 40%

Q30: Installed fasteners must meet what guidelines?

A: Engineering Requirements

Q31: Name of the guage used to inspect diameter of a rivet head?

A: Spoon guage

Q32: What is the typical pressure (pounds per square inch) for aviation pneumatic tools?

A: 90 psi

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg